



Quantum computing

Taras Holotyak

Content of course

- Theoretical basis for quantum computing.
- Practical basis for quantum computing.
- Near-term potential of quantum computing.
- Other applications of quantum information.

Quantum computing basics

Starting point

Quantum information theory is a generalization of **classical probability theory** where probabilities can be **negative**, or even **complex numbers**.

Quantum computing basics

Qubit definition

A **bit** is a classical system with *two* distinguishable states $|0\rangle$, $|1\rangle$, also called *Classical states*, or *standard basis states*.

A **qubit** (quantum bit) can be in a **superposition** of the classical states $|0\rangle$, $|1\rangle$.



Quantum computing basics

Qubit definition

Mathematically, states of a qubit are **complex linear combination** of $|0\rangle, |1\rangle$:

$$\begin{aligned} |\psi\rangle &= \alpha |0\rangle + \beta |1\rangle \\ &= \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} \alpha \\ \beta \end{pmatrix}. \end{aligned}$$

where $\alpha, \beta \in \mathbb{C}$ are complex numbers satisfying $|\alpha|^2 + |\beta|^2 = 1$.

In other words, $|\psi\rangle$ is a two-dimensional unit vector in $\mathbb{C}^2$.

Quantum computing basics

Qubit definition

Example: a qubit can be in the state

$$\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle .$$

$$\frac{1}{\sqrt{2}} |0\rangle - \frac{i}{\sqrt{2}} |1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} .$$

A non-valid qubit state:

$$i |0\rangle - \frac{1}{2} |1\rangle .$$

Quantum computing basics

Qubit definition

A qubit in the state $\alpha |0\rangle + \beta |1\rangle$ is commonly said to be $|0\rangle$ and $|1\rangle$ “at the same time”. But what does that mean?



α, β are like probabilities, except they can be *negative* or even *complex* numbers!

α, β are called the **amplitudes** of the states $|0\rangle$ and $|1\rangle$, respectively.

Quantum computing basics

Qubit definition

Classical Bit vs. Quantum Bit

CLASSICAL BITS:

- can be in two distinct states, 0 and 1
- can be measured completely
- are not changed by measurement
- can be copied
- can be erased

QUANTUM BITS:

- can be in state $|0\rangle$ or in state $|1\rangle$ or in any other state that is a linear combination of the two states
- can be measured partially with given probability
- are changed by measurement
- cannot be copied
- cannot be erased

Quantum computing basics

Advantages of Qubits & Enormous Quantum Power

- Adding qubits increases storage exponentially
 - Quantum computer doubles the power with every added qubit
 - To double the power of a digital computer 32bits -> 64 bits
 - To double the power of a quantum computer 32qubits -> 33 qubits
- Can do operations on all superpositions...like massively parallel computation
 - One math operation on 2^n numbers encoded in classical computers with n bits requires 2^n steps or parallel processors, but the same operation on 2^n numbers encoded by n qubits takes 1 step
 - A 64-bit computer can perform manipulation on 64-bit binary numbers at a time.
 - A 64-qubit quantum computer operates in a space of 2^{64} dimensions, or roughly 16,000,000,000,000,000,000 ($16 \cdot 10^{18}$) numbers to specify the state of the quantum system.
- This makes complex problems much easier to solve by quantum computer

Quantum computing basics

Qubit definition

The **dual/Hermitian conjugate** of column vectors (i.e., row vectors) are called “bra vectors”:

$$\langle 0| = (1, \ 0) \qquad \langle 1| = (0, \ 1)$$

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \alpha |0\rangle + \beta |1\rangle \qquad \langle\psi| = (\bar{\alpha} \ \bar{\beta}) = \bar{\alpha} \langle 0| + \bar{\beta} \langle 1|$$

$\bar{\alpha}, \bar{\beta}$ = **complex conjugates** of α, β .

Quantum computing basics

Qubit definition

Let $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$ and $|\theta\rangle = \gamma |0\rangle + \delta |1\rangle$ be column vectors.
Then we can take their inner product:

$$\langle\psi| |\theta\rangle = \langle\psi | \theta\rangle = \begin{pmatrix} \bar{\alpha}, & \bar{\beta} \end{pmatrix} \begin{pmatrix} \gamma \\ \delta \end{pmatrix} = \bar{\alpha}\gamma + \bar{\beta}\delta$$

Note the order of $|\psi\rangle$ and $|\theta\rangle$ matters!

$$\langle\theta | \psi\rangle = \bar{\gamma}\alpha + \bar{\delta}\beta = \overline{\langle\psi | \theta\rangle}$$

Quantum computing basics

Qubit definition

More generally, if

$$|\psi\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle + \cdots + \alpha_{d-1} |d-1\rangle$$

$$|\theta\rangle = \beta_0 |0\rangle + \beta_1 |1\rangle + \cdots + \beta_{d-1} |d-1\rangle$$

are d -dimensional column vectors, then their inner product is

$$\langle\psi | \theta\rangle = \overline{\alpha}_0\beta_0 + \cdots + \overline{\alpha}_{d-1}\beta_{d-1}$$

Dirac notation is very useful for quickly identifying scalars, row and column vectors, matrices in complicated expressions.

Quantum computing basics

Composite quantum systems

If qubit 1 is in state $|\psi_1\rangle$, and qubit 2 is in state $|\psi_2\rangle$, then the joint state of both qubits is

$$|\psi_1\rangle \otimes |\psi_2\rangle$$

This is a **four-dimensional** unit vector in the tensor product space $\mathbb{C}^2 \otimes \mathbb{C}^2$.

Quantum computing basics

Composite quantum systems

Quantum states live in a **Hilbert space** H . There's a precise mathematical definition, but for our purposes the important aspects of Hilbert spaces are:

- Finite-dimensional Hilbert spaces are isomorphic to $\mathbb{C}^d$ for some dimension d .
- There is an *inner product* operation between two vectors.
- You can take *tensor products* of Hilbert spaces.

The Hilbert space for two qubits is $\mathbb{C}^2 \otimes \mathbb{C}^2$, with basis $|0\rangle \otimes |0\rangle, |0\rangle \otimes |1\rangle, |1\rangle \otimes |0\rangle, |1\rangle \otimes |1\rangle$.

We can abbreviate this to

$$|0, 0\rangle \quad |0, 1\rangle \quad |1, 0\rangle \quad |1, 1\rangle$$

Quantum computing basics

Composite quantum systems

As well as scalar multiplication and matrix multiplication, to describe quantum computation we must consider a third form of multiplication on matrices, *tensor multiplication*. Let A and B be matrices of any dimension:

$$A \otimes B = \begin{bmatrix} a_{11}B & \dots & a_{1m}B \\ \vdots & \ddots & \vdots \\ a_{n1}B & \dots & a_{nm}B \end{bmatrix}$$

where $\otimes$ denotes the tensor product. For example:

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \otimes \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 4 & 6 \end{bmatrix}$$

In general if A is $n \times m$ and B is $n' \times m'$ then $A \otimes B$ is $nn' \times mm'$.

The tensor product is associative, so $A \otimes (B \otimes C) = (A \otimes B) \otimes C$.

Quantum computing basics

Composite quantum systems

Not all states in $\mathbb{C}^2 \otimes \mathbb{C}^2$ can be written as $|\psi_1\rangle \otimes |\psi_2\rangle$ (called **product states**).

An example of an **entangled state**, called **Einstein-Podolsky-Rosen (EPR) pair** or **Bell pair**.

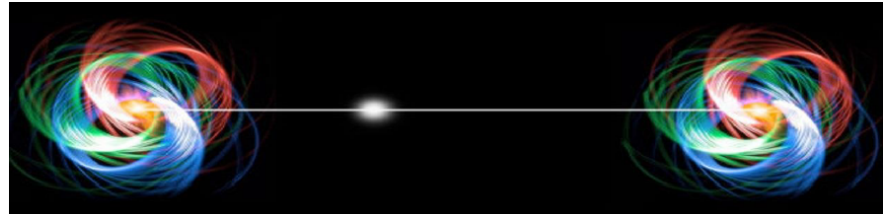
$$|\Phi\rangle = \frac{1}{\sqrt{2}} \left(|0, 0\rangle + |1, 1\rangle \right)$$

Entanglement is a **uniquely quantum phenomenon**: one qubit is intimately linked to another qubit in a way that classical physics cannot explain.

Quantum computing basics

Composite quantum systems

Entanglement

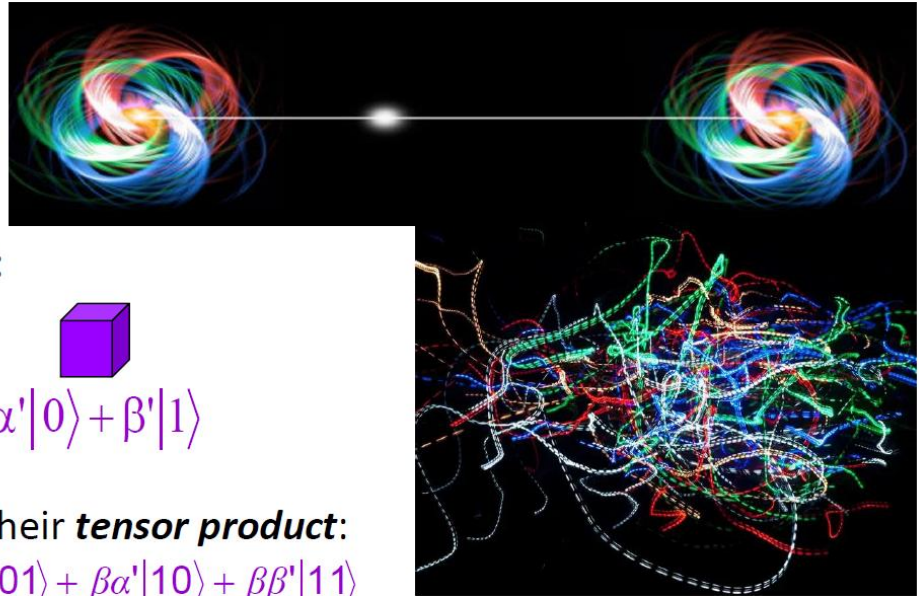


- When a pair or group of particles is generated, interact, or share spatial proximity in a way such that the quantum state of each particle of the pair or group cannot be described independently of the state of the others, **even when the particles are separated by a large distance**.
- An entangled pair is a single quantum system in a superposition of equally possible states. The entangled state contains no information about the individual particles, only that they are in opposite states.
- If the state of one is changed, the state of the other is **instantly** adjusted to be consistent with quantum mechanical rules.
- If a measurement is made on one, the other will **automatically** collapse.
- Quantum entanglement is at the heart of the disparity between classical and quantum physics: ***entanglement is a primary feature of quantum mechanics lacking in classical mechanics.***
- Entanglement is a joint characteristic of two or more quantum particles.
- Einstein called it “**spooky actions at a distance**”

Quantum computing basics

Composite quantum systems

Entanglement



Suppose that two qubits are in states:


 $\alpha|0\rangle + \beta|1\rangle$


 $\alpha'|0\rangle + \beta'|1\rangle$

The state of the combined system is their **tensor product**:

$$(\alpha|0\rangle + \beta|1\rangle)(\alpha'|0\rangle + \beta'|1\rangle) = \alpha\alpha'|00\rangle + \alpha\beta'|01\rangle + \beta\alpha'|10\rangle + \beta\beta'|11\rangle$$

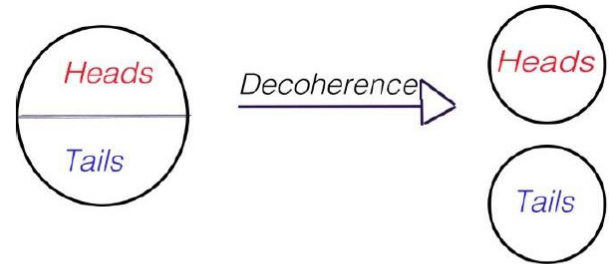
Question: what are the states of the individual qubits for

1. $\frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle - \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle$ an **independent** state
2. $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$ an **entangled** state

Quantum computing basics

Observing qubits

Decoherence



- Quantum decoherence is the loss of superposition, because of the spontaneous interaction between a quantum system and its environment.
- Decoherence can be viewed as the loss of information from a system into the environment.
- The reason why quantum computers still have a long way to go because superposition and entanglement are extremely fragile states.
- Preventing decoherence remains the biggest challenge in building quantum computers.

Quantum computing basics

Observing qubits

The state of a qubit cannot be directly observed. It must be **measured**, yielding a classical state $|0\rangle$ or $|1\rangle$ with probabilities

$$\Pr[\text{observing } |0\rangle] = |\alpha|^2 \qquad \Pr[\text{observing } |1\rangle] = |\beta|^2.$$

Because qubit states have unit length, these probabilities add up to 1.

This formula is called the **Born Rule**.

Quantum computing basics

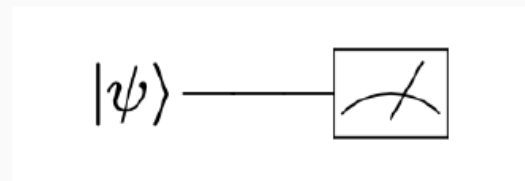
Observing qubits

After measurement, the system becomes **classical**.

The state of qubit **collapses** to either $|0\rangle$ or $|1\rangle$, and the previous state is lost.

In quantum mechanics, measurement generally disturbs the system.

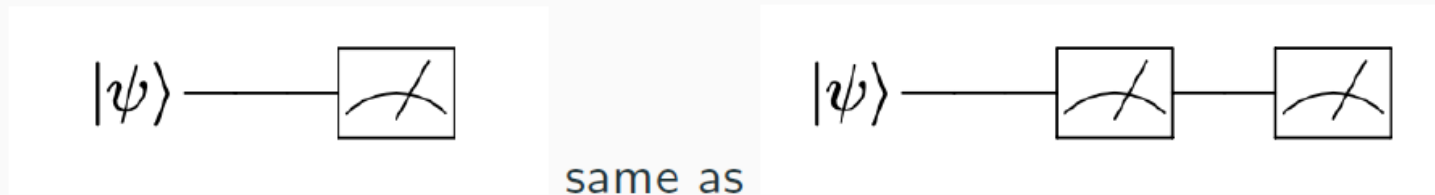
We represent qubit measurements using this diagram:



Quantum computing basics

Observing qubits

If the state collapses to the classical state $|0\rangle$ and we measure it again, it stays in state $|0\rangle$ **with probability 1**. Same with collapsing to $|1\rangle$.



Measuring a system twice is the same as measuring once.

Quantum computing basics

Schrödinger's cat



A box with two classical states:

sleeping cat: $|0\rangle$

awake cat: $|1\rangle$

In quantum mechanics, the box can be in a superposition of sleeping and awake cat, *as long as you don't open the box* (i.e. measure it).

Quantum computing basics

Schrödinger's cat



Suppose the box starts in the state $\frac{1}{\sqrt{3}} |0\rangle - i \frac{\sqrt{2}}{\sqrt{3}} |1\rangle$ and is measured.

1. With probability $\left| \frac{1}{\sqrt{3}} \right|^2 = \frac{1}{3}$, the state collapses to $|0\rangle$ (i.e., sleeping cat).
2. With probability $\left| -i \frac{\sqrt{2}}{\sqrt{3}} \right|^2 = \frac{2}{3}$, the state collapses to $|1\rangle$ (i.e., awake cat).

Quantum computing basics

Transformations on qubits

- A **quantum gate** (or **quantum logic gate**) is a basic quantum circuit operating on qubits.
- They are the building blocks of quantum circuits, like classical logic gates are for conventional digital circuits
- Due to the normalization condition every gate operation U has to be unitary: $UU^* = I$
- The number of qubits in the input and output of the gate must be equal; a gate which acts on n qubits is represented by $2^n \times 2^n$ unitary matrix
- Unlike many classical logic gates, quantum gates are reversible.

Quantum computing basics

Transformations on qubits

In addition to measurement, the state of a qubit can change via a **unitary transformation**. Just like transformations in classical reversible computing, unitary transformations can be represented as matrices.

We represent a unitary transform U acting on state $|\psi\rangle$ using the following circuit diagram:



Quantum computing basics

Several equivalent definitions of unitary matrices

Definition 1. The inverse of U is its Hermitian conjugate $U^\dagger$, pronounced “ U dagger”, whose (i, j) ’th entry is the *complex conjugate* of the (j, i) ’th entry of U :

$$U_{i,j}^\dagger = \overline{U_{j,i}}$$

Definition 2. U maps unit vectors to unit vectors (i.e. quantum states to quantum states)

Definition 3. The rows of U form an orthonormal basis for $\mathbb{C}^d$, and the columns form an orthonormal basis for $\mathbb{C}^d$.

Definition 4. U preserves the inner products between vectors: inner product between $|\psi\rangle$ and $|\theta\rangle$ is the same as the inner product between $U|\psi\rangle$ and $U|\theta\rangle$.

Quantum computing basics

Examples of qubit unitary matrices

Single Qubit Gate

$$|0\rangle \text{ --- } \boxed{U} \text{ --- Any state } |\psi\rangle$$

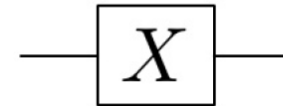
- **Pauli-X gate**

$$|0\rangle \rightarrow |1\rangle, \quad |1\rangle \rightarrow |0\rangle$$

Dirac notation

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Matrix representation



Circuit representation

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$



$$X \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \cdot 1 + 1 \cdot 0 \\ 1 \cdot 1 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



$$X \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \cdot 0 + 1 \cdot 1 \\ 1 \cdot 0 + 0 \cdot 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

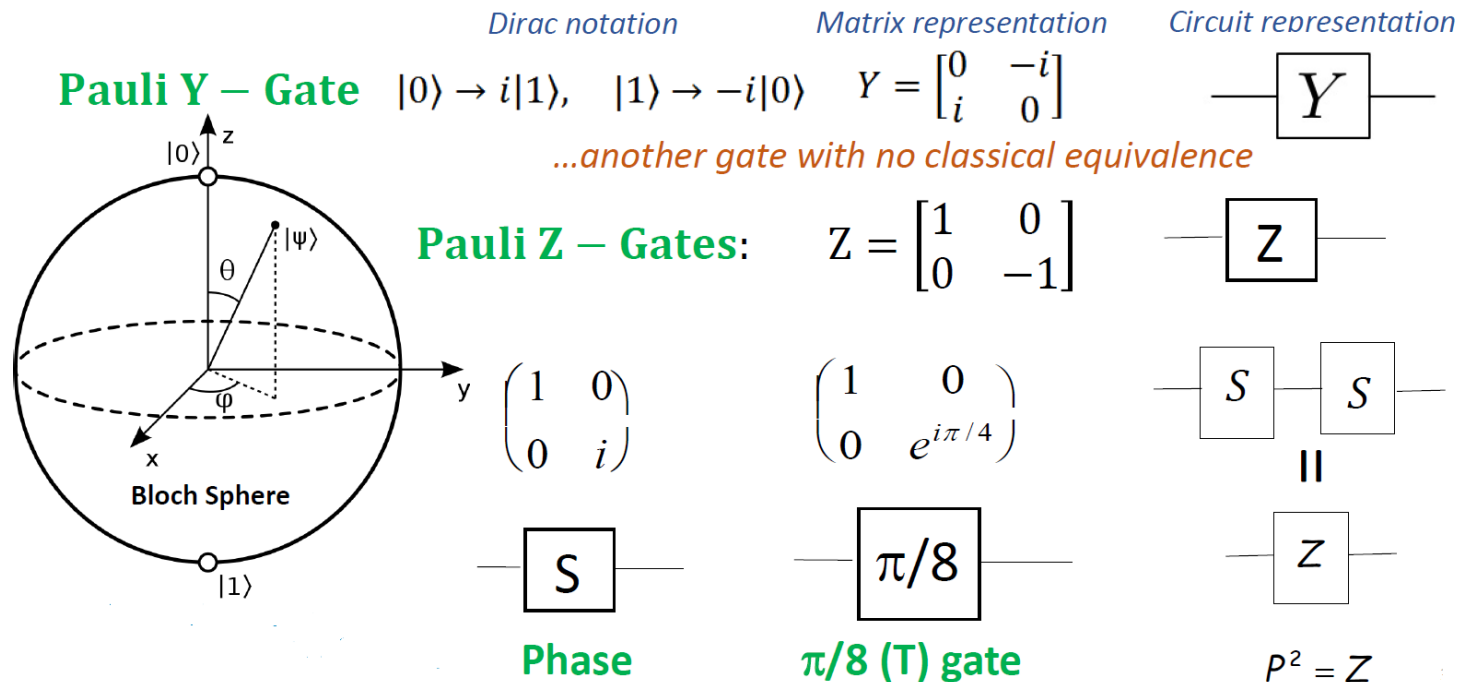
- Acting on pure states becomes a **classical NOT** gate

$$|0\rangle \text{ --- } \boxed{X} \text{ --- } |1\rangle$$

Quantum computing basics

Examples of qubit unitary matrices

Single Qubit Gates



Quantum computing basics

Examples of qubit unitary matrices

Hadamard Gate

- Acts on a single qubit
 - Corresponding to the Hadamard transform leading to *superposition*

Dirac notation

$$|0\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$|1\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

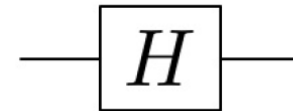
$$(|0\rangle + |1\rangle)/\sqrt{2} \rightarrow |0\rangle$$

$$(|0\rangle - |1\rangle)/\sqrt{2} \rightarrow |1\rangle$$

Unitary matrix

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Circuit representation



no classical equivalent!

Quantum computing basics

Examples of qubit unitary matrices

The Amazing H-Gate

- After a qubit in state $|0\rangle$ or $|1\rangle$ has been acted upon by a H gate, the state of the qubit is an equal superposition of $|0\rangle$ and $|1\rangle$. Thus, the qubit goes from a deterministic state to a truly random state, i.e., if the qubit is now measured, we will measure $|0\rangle$ or $|1\rangle$ with equal probability.
- We see that H is its own inverse, that is, $H^{-1} = H$ or $H^2 = I$. Therefore, by applying H twice to a qubit we change nothing. [This is amazing!](#)
- By applying a randomizing operation to a random state produces a deterministic outcome!
- **One of the most important gates in quantum computing!**

Quantum computing basics

Examples of qubit unitary matrices

CNOT Gate

- **Controlled NOT** gate

$$|00\rangle \rightarrow |00\rangle, \quad |01\rangle \rightarrow |01\rangle$$

- Acts on two qubits

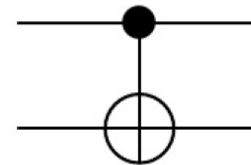
$$|10\rangle \rightarrow |11\rangle, \quad |11\rangle \rightarrow |10\rangle$$

- If the control qubit is set to 0, target qubit is the same
- If the control qubit is set to 1, target qubit is flipped

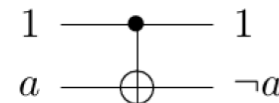
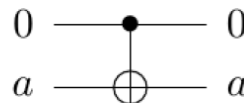
Matrix representation

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Circuit representation



- Equivalent to classical gate operation **XOR**



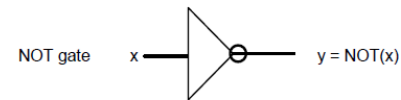
The output is “true” if and only if exactly one of the operands has a value of “true”.

Quantum computing basics

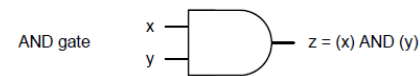
Examples of qubit unitary matrices

Quantum vs. Classic Gates

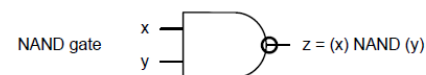
Operator	Gate(s)	Matrix
Pauli-X (X)		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)		$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Controlled Z (CZ)		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
SWAP		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Toffoli (CCNOT, CCX, TOFF)		$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$



x	y
0	1
1	0



x	y	z
0	0	0
0	1	0
1	0	0
1	1	1



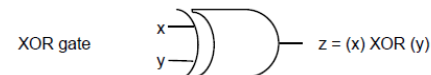
x	y	z
0	0	1
0	1	1
1	0	1
1	1	0



x	y	z
0	0	0
0	1	1
1	0	1
1	1	1



x	y	z
0	0	1
0	1	0
1	0	0
1	1	0



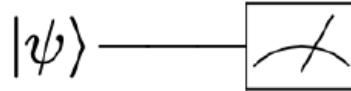
x	y	z
0	0	0
0	1	1
1	0	1
1	1	0

Quantum computing basics

Qubit processing: quantum vs classical bits

Suppose we were physically handed a qubit (say Schrödinger's box) whose state $|\psi\rangle$ was either $|+\rangle$ or $|-\rangle$. Is there a way we can tell the difference?

Opening the box (i.e., measuring) would yield a sleeping or awake cat with equal probability in both cases.



Quantum computing basics

Qubit processing : quantum vs classical bits



Solution: Apply H to qubit before measuring!

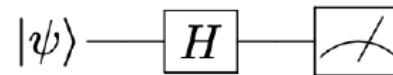
Case 1: $|\psi\rangle = |+\rangle$. Applying H , we get

$$H |+\rangle = |0\rangle$$

Measuring yields $|0\rangle$ all the time!

Quantum computing basics

Qubit processing: quantum vs classical bits



Solution: Apply H to qubit before measuring!

Case 2: $|\psi\rangle = |-\rangle$. Applying H , we get

$$H|-\rangle = |1\rangle$$

Measuring yields $|1\rangle$ all the time!

Quantum computing basics

Qubit processing: quantum vs classical bits

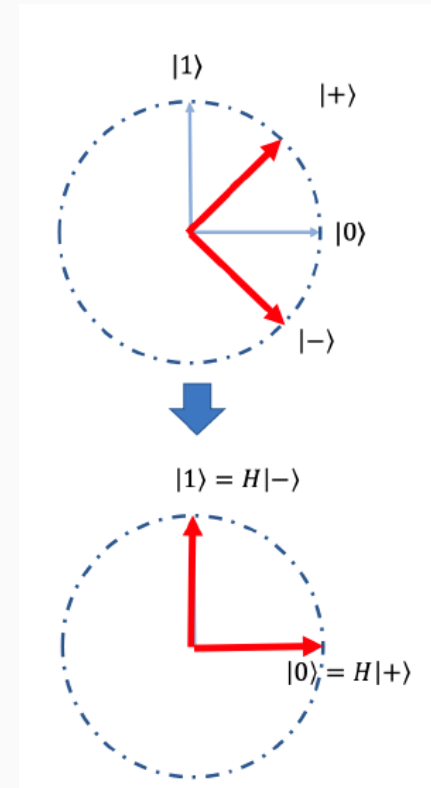
The states

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

and

$$|-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

form an orthonormal basis for $\mathbb{C}^2$.

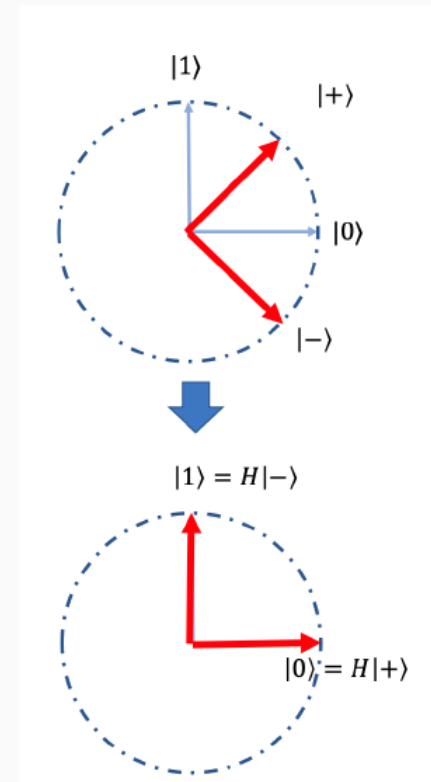


Quantum computing basics

Qubit processing: quantum vs classical bits

In quantum mechanics, orthogonal states can be **perfectly distinguished** by applying an appropriate unitary matrix and then measuring in the standard basis.

The Hadamard matrix maps the $\{ |+\rangle, |-\rangle \}$ basis to the standard $\{ |0\rangle, |1\rangle \}$ basis (and vice versa).



Quantum computing basics

Qubit processing: quantum vs classical bits

The Hadamard gate turns a classical state into a quantum superposition:

$$\begin{aligned}|+\rangle &= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = H|0\rangle \\ |-\rangle &= \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = H|1\rangle .\end{aligned}$$

The measurement statistics of $|+\rangle$, $|-\rangle$ look the same, so can these states be distinguished?

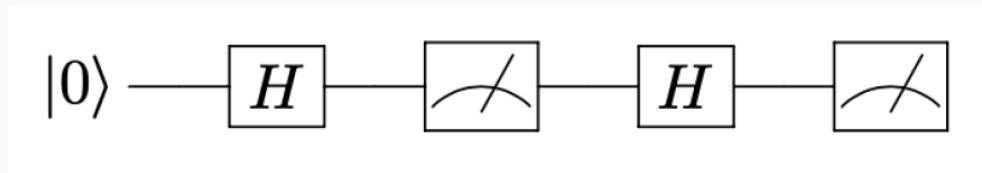
Yes, by first applying a unitary, namely, the Hadamard matrix again!

$$|0\rangle = H|+\rangle \quad |1\rangle = H|-\rangle .$$

Quantum computing basics

Qubit processing: quantum vs classical bits

Consider **Experiment A**:



Final distribution of outcomes: $|0\rangle$, $|1\rangle$ with equal probability.

Based on this, one might conclude that the Hadamard gate simply creates a random classical bit!

Quantum computing basics

Qubit processing: quantum vs classical bits

Consider **Experiment B**:

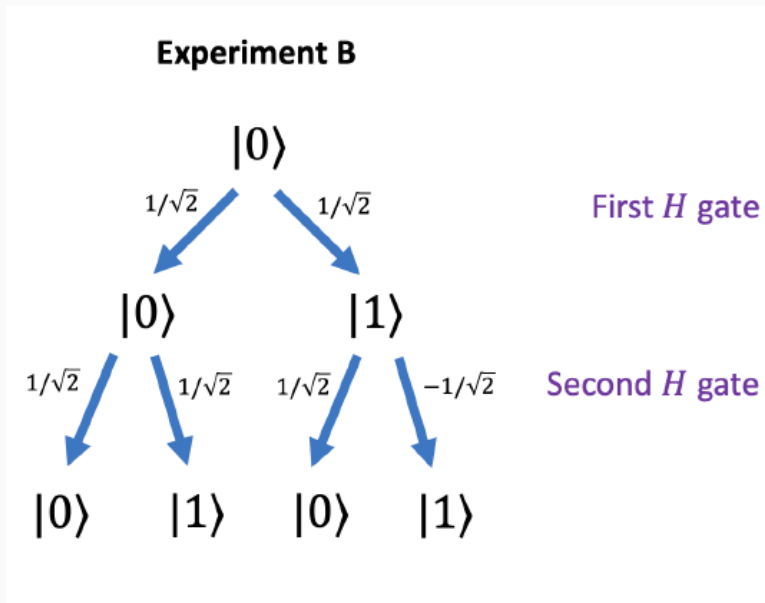


Final distribution of outcomes: $|0\rangle$ always.

Inserting or removing intermediate measurement can drastically change the behavior of a quantum system!.

Quantum computing basics

Qubit processing: quantum interference

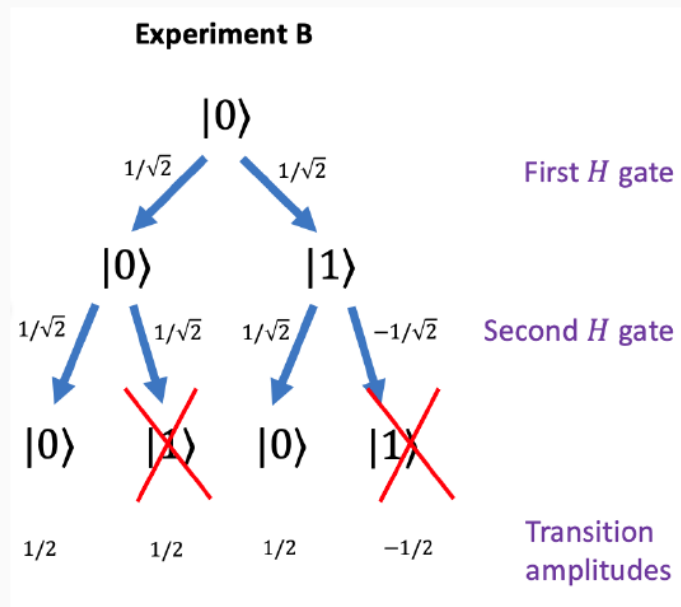


Applying multiple unitary operations to a qubit gives rise to a tree of “paths” between states.

An edge between state $|a\rangle$ to $|b\rangle$ has a transition amplitude associated with, depending on the unitary.

Quantum computing basics

Qubit processing: quantum interference



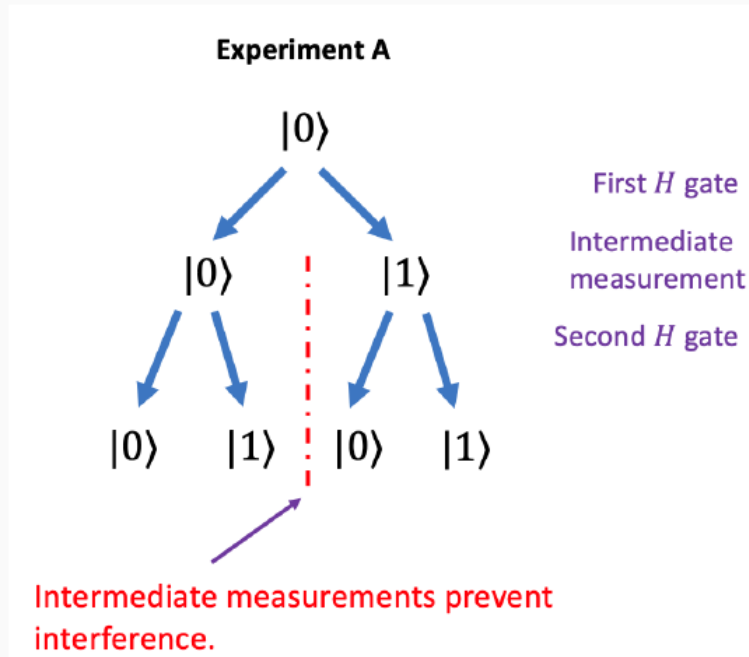
Each path has associated transition amplitude: product of amplitudes along the path.

Before measurement, amplitude of ending up in state $|b\rangle$ is the sum of transition amplitudes of all paths that end at $|b\rangle$.

Since transition amplitudes can be negative or even complex, the paths can constructively or destructively interfere with each other!

Quantum computing basics

Qubit processing: quantum interference



If there is an intermediate measurement, only paths consistent with the measurement outcome are added up.

Takeway: Intermediate measurements destroy superpositions and prevent interference!

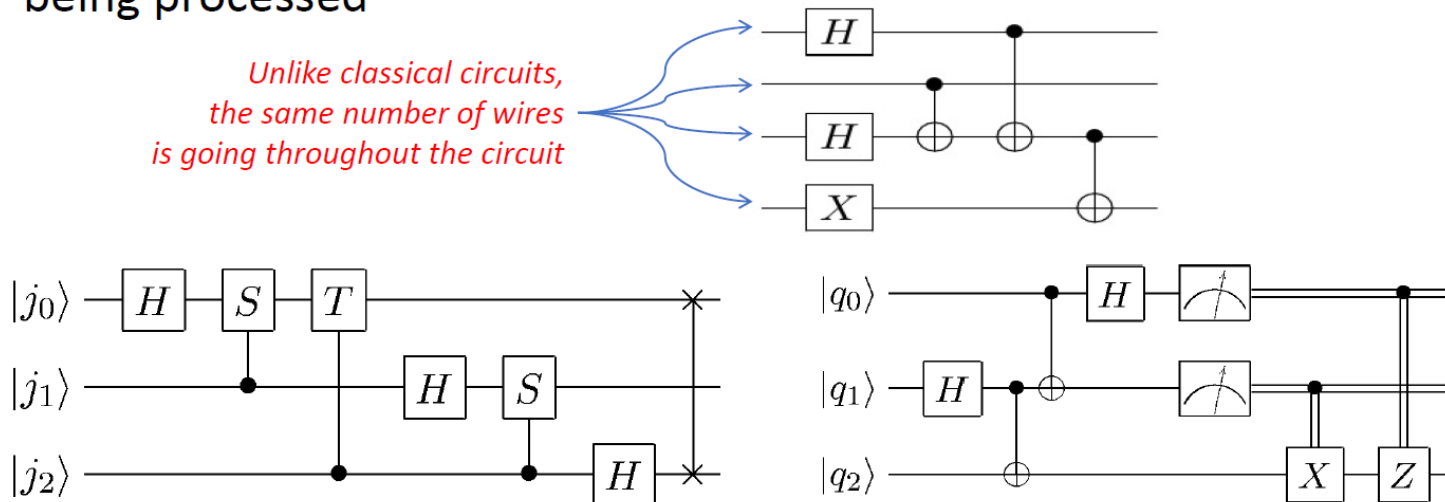
In classical physics, we only add up probabilities, which are positive. In quantum physics, we add up amplitudes, which can be negative. Interference is an example of "quantum weirdness".

Quantum computing basics

Qubit processing: quantum circuits

- A quantum circuit is a model for quantum computation in which a computation is a **sequence** of **quantum gates** with n -qubit register linked by “wires”
- The circuit has fixed “width” corresponding to the number of qubits being processed

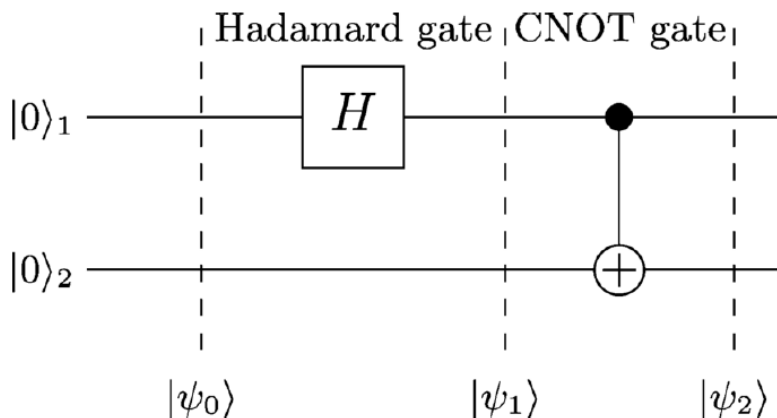
*Unlike classical circuits,
the same number of wires
is going throughout the circuit*



Quantum computing basics

Qubit processing: quantum circuits

How to generate entanglement?



A simplified quantum circuit diagram showing two qubits, both initialized to $|0\rangle$. The top qubit passes through a Hadamard gate (H) and then acts as the control for a CNOT gate on the bottom qubit, which is the target. The final state is shown in curly braces as $\frac{|00\rangle + |11\rangle}{\sqrt{2}}$.

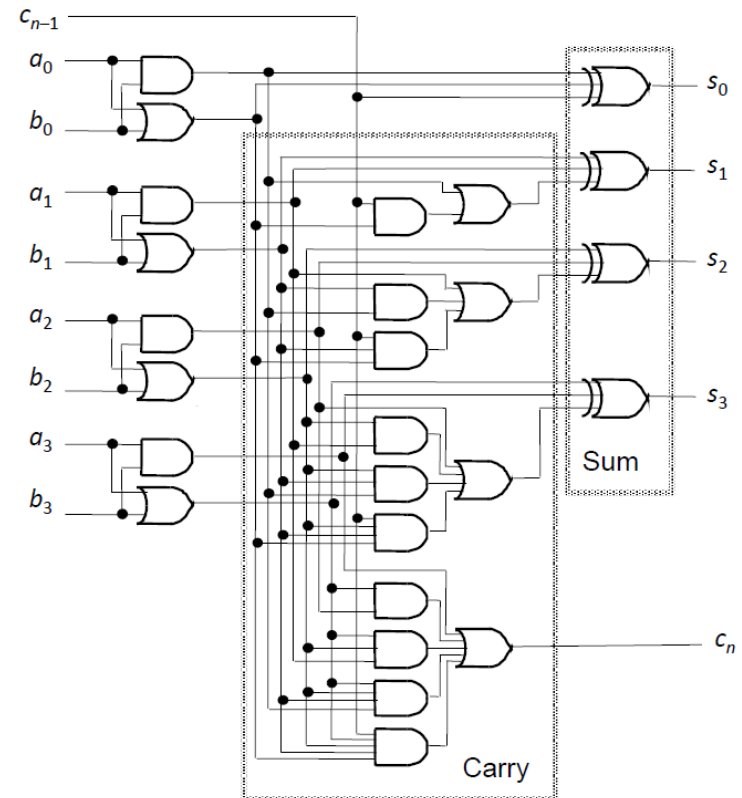
Using two qubits to generate an entanglement state, also called Bell state,
with a Hadamard gate and a CNOT gate

Quantum computing basics

Qubit processing: quantum vs classical circuits

- Classical Logic Circuits

- Circuit behavior is governed implicitly by **classical physics**
- Signal states are simple bit vectors, e.g. $X = 01010111$
- Operations are defined by Boolean Algebra
- No restrictions exist on copying or measuring signals
- Small well-defined sets of universal gate types, e.g. {NAND}, {AND,OR,NOT}, {AND,NOT}, etc.
- Well developed CAD methodologies exist
- Circuits are easily implemented in fast, scalable and macroscopic technologies such as CMOS



Quantum computing basics

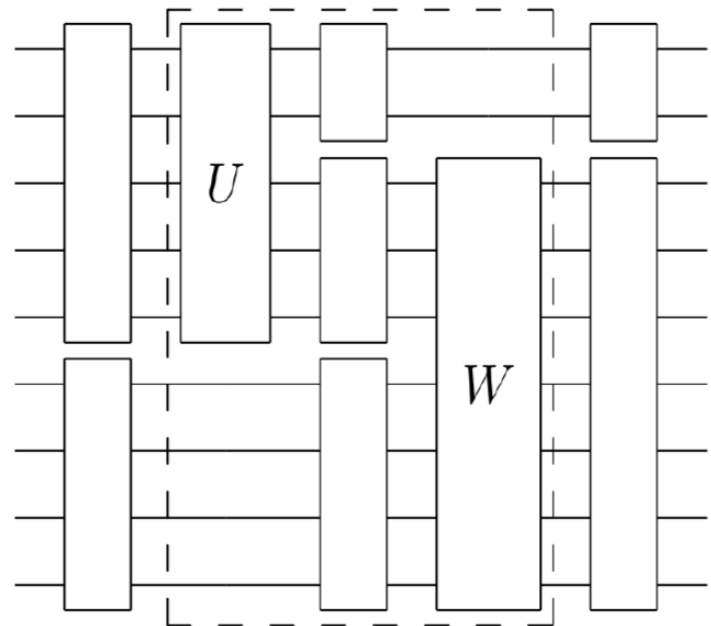
Qubit processing: quantum vs classical circuits

- Quantum Logic Circuits

- Circuit behavior is governed explicitly by **quantum mechanics**
- Signal states are vectors interpreted as a **superposition** of binary “qubit” vectors with complex-number coefficients

$$|\Psi\rangle = \sum_{i=0}^{2^n-1} c_i |i_{n-1}i_{n-2}\dots i_0\rangle$$

- Operations are defined by linear algebra over Hilbert Space and can be represented by **unitary matrices** with complex elements
- Severe restrictions exist on **copying** and **measuring** signals
- Many universal gate sets exist but the best types are not obvious
- Circuits must use microscopic technologies that are slow, fragile, and not yet scalable, e.g., NMR



Quantum computing basics

Quantum algorithms

- It may be possible to solve a problem on a quantum system much faster (i.e., using fewer steps) than on a classical computer
- Factoring and searching are examples of problems where quantum algorithms are known and are faster than any classical ones
- Implications for cryptography, information security
- What makes a quantum algorithm potentially faster than any classical one?
 - **Quantum parallelism:** by using superpositions of quantum states, the computer is executing the algorithm on all possible inputs at once
 - **Dimension of quantum Hilbert space:** the “size” of the state space for the quantum system is exponentially larger than the corresponding classical system
 - **Entanglement capability:** different subsystems (qubits) in a quantum computer become entangled, exhibiting nonclassical correlations

Quantum computing basics

Quantum algorithms

Algorithms	Classical steps	quantum logic steps
Fourier transform e.g.: - Shor's prime factorization - discrete logarithm problem - Deutsch Jozsa algorithm	$N \log(N) = n 2^n$ $N = 2^n$ - n qubits - N numbers	$\log^2(N) = n^2$ - hidden information! - Wave function collapse prevents us from directly accessing the information
Search Algorithms	N	$\sqrt{N}$
Quantum Simulation	c^N bits	kn qubits

Quantum computing basics

Quantum programming

- **QCL** (Quantum Computation Language)

```
/* Remove "/" if starting interpreter with -n option */  
// extern operator H(qreg q);
```

C-like syntax

```
procedure FlipCoin() {  
    qureg q[1]; int x;
```

```
    reset;
```

```
    H(q);
```

```
    measure q, x;
```

```
    if x == 1 { print "Heads"; }
```

```
    if x == 0 { print "Tails"; }
```

```
    reset;
```

```
}
```

*allows combining of
quantum and
classical code*



<http://tph.tuwien.ac.at/~oemer/qcl.html>

Quantum computing basics

Quantum programming



- **QISKit** (Quantum Information Science Kit)

Qiskit is an open-source framework for quantum computing. It provides tools for creating and manipulating quantum programs and running them on prototype quantum devices on IBM Quantum Experience over **Cloud-based access**



```
In [7]: from qiskit import QuantumRegister, ClassicalRegister, QuantumCircuit
        from qiskit.tools.visualization import circuit_drawer
        import numpy as np

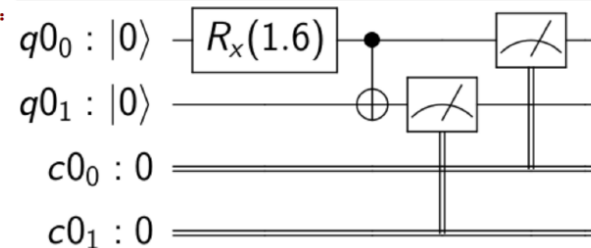
        qr = QuantumRegister(2)
        cr = ClassicalRegister(2)
        qp = QuantumCircuit(qr, cr)

        qp.rx( np.pi/2, qr[0])
        qp.cx(qr[0], qr[1])

        qp.measure(qr, cr)

        circuit_drawer(qp)
```

Out[7]:



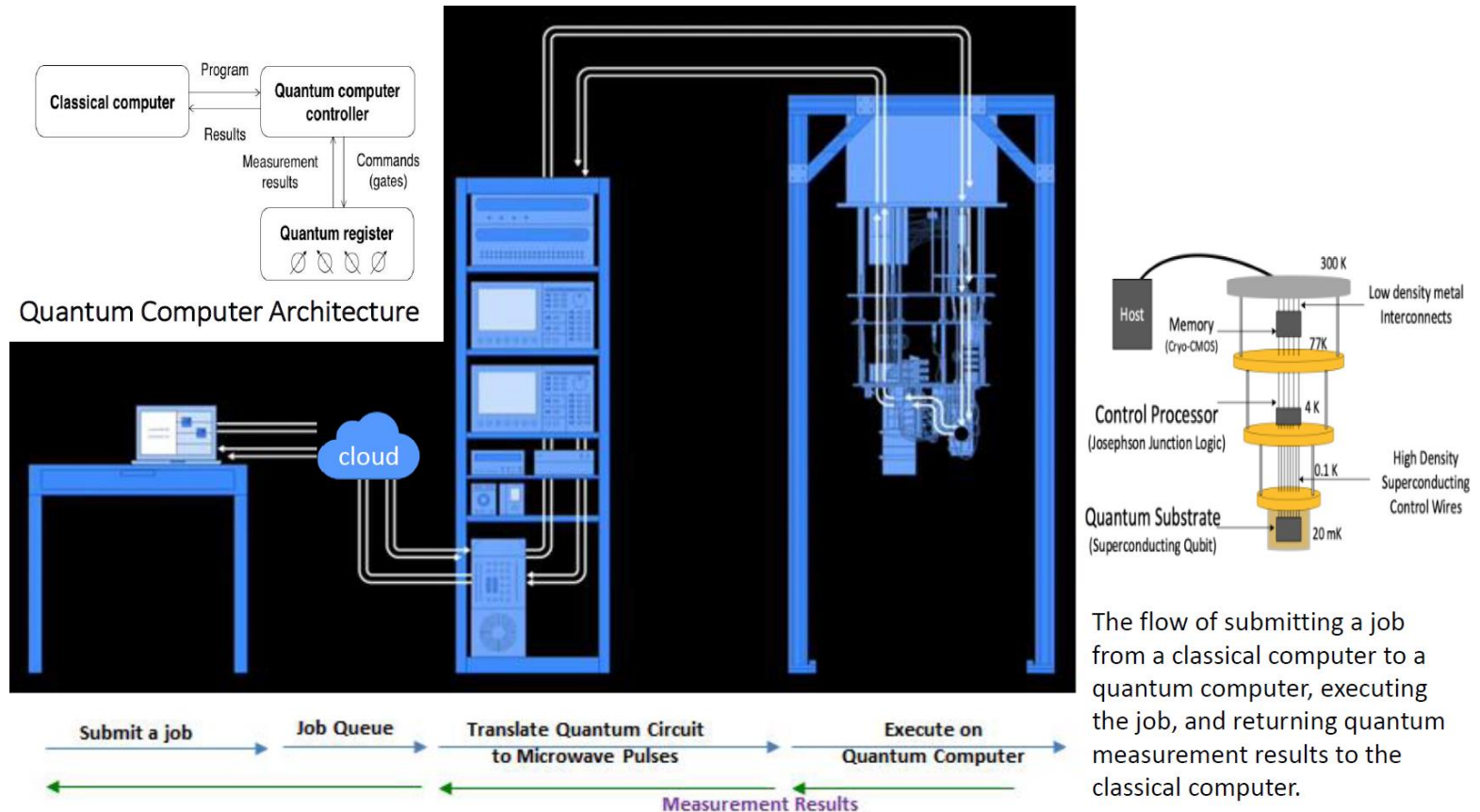
```
In [1]: from qiskit import *

In [2]: qr = QuantumRegister(2)
        cr = ClassicalRegister(2)

In [3]: c = QuantumCircuit(qr, cr) # c = QuantumCircuit(2,2)
```

Quantum computing basics

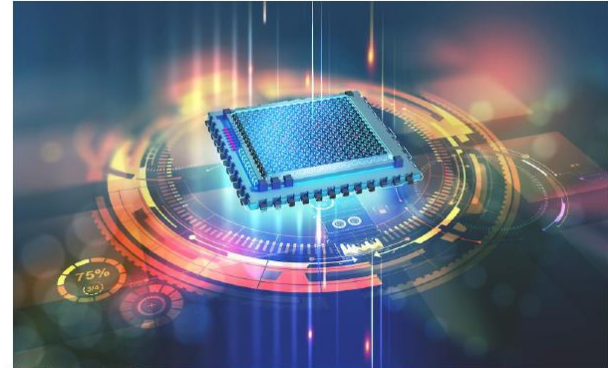
Quantum computer architecture



Quantum computing basics

Quantum computer architecture

- Requirements
 - Qubit implementation itself;
 - Control of unitary evolution;
 - Initial state preparation (qubits);
 - Measurement of the final state(s).
- Other Considerations
 - Systems have to be almost completely isolated from their environment
 - The coherent quantum state has to be preserved
 - Decoherence times have to be very long
 - Performing operations on several qubits in parallel
 - Physical system with two uniquely addressable states
 - A universal set of quantum gates
 - Ability to entangle two qubits



Quantum computing basics

Quantum volume

- **Quantum volume** V_Q is a metric that measures the capabilities and error rates of a quantum computer.
- Defined by Nikolaj Moll *et al.* *Quantum Sci. Technol.* **3**, 030503 (2018).
- It depends on the **number** of qubits N as well as the number of steps that can be executed, the circuit **depth** d : $\tilde{V}_Q = \min[N, d(N)]^2$.
- IBMs modified the quantum volume definition: $\log_2 V_Q = \arg \max_{n \leq N} \{\min [n, d(n)]\}$

Date	Quantum volume (circuit size)	Manufacturer	Notes
2020, January	32 (5×5)	IBM	"Raleigh" (28 qubits)
2020, June	64 (6×6)	Honeywell	6 qubits
2020, August	64 (6×6)	IBM	27 qubits
2020, November	128 (7×7)	Honeywell	"H1" (10 qubits)